SCIENCE FAIR PROJECT GUIDE

How Fast Do Drawings Need to Move to Become a Movie?

Testing frame rate, persistence of vision, and the phi phenomenon with a hand-drawn car chase

Eyes & vision science | Movies & animation | Engineering (zoetrope build) | Ages 9–11



Separate drawings, each one a little different...

...your brain sees motion

THE BIG QUESTION

What is the fewest frames per second (FPS) before drawings stop looking like separate pictures and start looking like a movie?

SCIENTIST

Name: _____

Grade: _____

School: _____

Fair date: _____

WHAT'S INSIDE

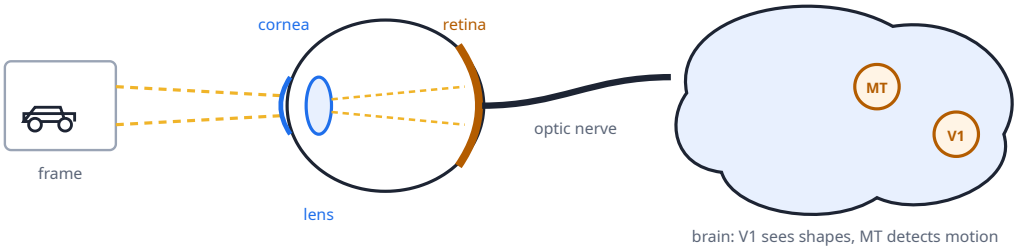
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BACKGROUND RESEARCH

How do your eyes and brain see motion?

A movie is just a lot of **still pictures** shown very quickly, one after another. Each still picture is called a **frame**. How many frames are shown each second is the **frame rate**, measured in **frames per second (FPS)**. Somewhere along the way your brain stops seeing a slideshow and starts seeing motion. That's the moment this project is trying to find.

The trip from picture to brain



1. **Cornea and lens** bend light to focus the picture.
2. The **retina** at the back of the eye has about 120 million **rods** (for dim light and movement) and 6 million **cones** (for color and detail). They turn light into electrical signals.
3. The **optic nerve** carries those signals to the back of the brain. Area **V1** sees shapes and edges, and an area called **MT** works out what is moving and which way.

Persistence of vision

After a picture disappears, your eye and brain keep a fading trace of it for a split second, roughly 1/15 to 1/10 of a second. If the next picture shows up before the trace fades, you don't notice the gap. Try it: stare at a bright window, then close your eyes and you'll see an afterimage.

The phi phenomenon

When two pictures flash one after the other in slightly different places, your brain *fills in* the movement between them, even though nothing actually moved. Scientists call this **apparent motion**. It's the same reason a row of blinking lights looks like it's "running."

SCIENTIST'S NOTE

For about 100 years, people said "persistence of vision" was the whole reason movies work. Today scientists think the brain's **motion detectors** (the phi phenomenon) do most of the work. Persistence of vision mostly explains why you don't see the dark gaps *between* frames. Explaining the difference will impress the judges.

Frame rates in real life

Where you see it	Typical FPS
Old hand-cranked flipbooks and early silent films	about 12–18
Hand-drawn cartoons (often drawn "on twos," so each drawing is shown twice)	12 drawings per second
Movies in theaters	24
TV and phone videos	30 or 60
Fast video games	60–144+

PLAN THE EXPERIMENT

Question, hypothesis and variables

TESTABLE QUESTION

At what frame rate (FPS) do most people stop seeing separate drawings and start seeing smooth motion?

My hypothesis

Use the form "If... then... because...". Write your own guess *before* testing anyone. A wrong guess is still good science!

Example: "If I show my car chase at different speeds, then most people will see smooth motion at about 12 FPS, because cartoons are often drawn at 12 drawings per second."

Variables

A fair test changes only **one** thing at a time.

Type	What it means	In this project
Independent variable (what I change)	The one thing you change on purpose	Frame rate: 2, 4, 6, 8, 10, 12, 15, 20, 24, 30 FPS
Dependent variables (what I measure)	What you watch or measure to see the effect	1. Each volunteer's answer: *"pictures"* or *"movie"* 2. Smoothness rating from 1 (choppy) to 5 (smooth)
Controlled variables (kept the same)	Everything else, kept the same so the test is fair	The same drawings, number of frames, screen or zoetrope, viewing distance (about 2 feet), room lighting, watching time (5 seconds per speed), and instructions read from a script

Rating scale (read this to every volunteer)

Score	What it looks like
1	A slideshow. I can clearly see each separate drawing.
2	Jumpy. I can tell something is moving, but it jerks.
3	In-between. Mostly moving, but still a bit choppy.
4	Almost smooth, with a small flicker or stutter.
5	Totally smooth, like a real cartoon.

KEY DEFINITION: THE "MOVIE THRESHOLD"

For each volunteer, the **movie threshold** is the *lowest* FPS where they answered "movie" *and* rated it 4 or 5. This is the number you'll average at the end.

GET READY

Materials list

There are two ways to show the animation. **Use both if you can.** The **digital flipbook** gives exact, fair FPS numbers for the experiment. The **zoetrope** is the eye-catching machine for the display table, and it shows how people watched "movies" in the 1800s.

For the drawings (everyone)

- Plain white paper or index cards
- Pencil, eraser and a black fine-tip marker
- Ruler
- Light source for tracing (a window or tablet screen)
- Colored pencils (optional)

Option A: Digital flipbook

- Phone or tablet to photograph the drawings
- A computer, tablet or phone to play them back
- An app that lets you set the FPS exactly, such as **Stop Motion Studio** (free) or **Scratch** (free, scratch.mit.edu)
- Or the **built-in FPS tester** in the online version of this guide

Option B: Zoetrope

- Black card stock (2–3 sheets) or a sturdy cardboard cake round
- A round base about 8–10 inches across (a cake pan, a pizza box circle or a paper plate stack)
- Scissors or a craft knife (adult use only)
- Tape, glue stick and a ruler
- A small piece of bright tape to mark one spot

Something to spin it (pick one):

- **Record turntable**: the easiest choice, with steady speeds of 33⅓, 45 and sometimes 78 RPM
- **Lego motor** (Technic, Powered Up or Spike) with gears to slow it down
- **Lazy Susan** spun by hand (fun, but the speed isn't steady)
- **Bike wheel** with the bike flipped upside down (adult supervision only, keep fingers away from the spokes)

For collecting data

- A stopwatch or phone timer
- Printed data sheets (page 9) and a clipboard
- At least **10 volunteers**. More is better: 15–20 makes the results more trustworthy. Try for a mix of ages: kids, teens, parents and grandparents.
- Index cards with the rating scale printed on them, so volunteers can point to a number

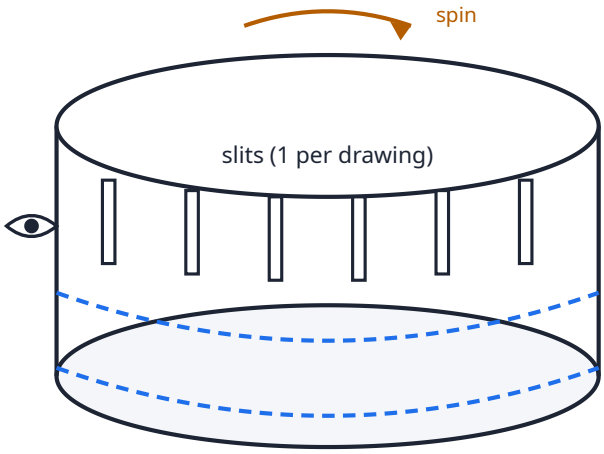
SAFETY

- An adult does all craft-knife cutting.
- Never touch a spinning wheel or turntable. Stop it first.
- Ask volunteers if flashing or flickering images bother them. Anyone with a history of seizures or migraines should skip the test.

ENGINEERING BUILD

Build a zoetrope

A **zoetrope** ("wheel of life") is a spinning drum with slits. You look *through* the slits at drawings on the inside wall. Each slit works like a camera shutter: you see one drawing for a split second, then darkness, then the next drawing. That flash-and-gap pattern is why the pictures don't blur together.



picture strip on the inside wall (between dashed lines)

Look through a slit (at eye level) at the drawing on the far side.

Build steps

1. **Pick the number of frames.** Use **12**. It divides evenly and is enough for a smooth loop.
2. **Measure the drum.** Circumference = 3.14 × diameter. For an 8-inch base, that's about 25 inches. Each drawing slot is then 25 ÷ 12 ≈ 2.1 inches wide.
3. **Make the wall.** Tape black card strips together into a band about 4 inches tall and as long as the circumference, plus 0.5 inch for overlap.
4. **Cut 12 slits** along the top inch of the band, evenly spaced (2.1 inches apart). Each slit is about ⅛ inch wide and 1 inch tall. *Narrower slits give a sharper image but a darker one.*
5. Glue the band around the base, with the slits at the top.
6. Put the **picture strip** (page 7) inside, along the bottom of the wall, with each drawing lined up *between* two slits.
7. Stick a piece of bright tape on the outside of the drum. You'll use it to count spins.

Zoetrope math: turning spins into FPS

FPS = drawings on the strip × spins per second

To find spins per second: count how many times the tape mark passes you in **10 seconds**, then divide by 10. Do it 3 times and take the average.

Spinner	Spins per minute (RPM)	Spins per second	FPS with 12 drawings	FPS with 6 drawings
Turntable at 33⅓	33.3	0.56	**6.7**	3.3
Turntable at 45	45	0.75	**9**	4.5
Turntable at 78	78	1.3	**15.6**	7.8
Lego motor (example)	120	2	**24**	12

Tip: you can get more speeds from one turntable by using **two strips**, one with 12 drawings and one with 6 (use every other drawing, and cut a band with 6 slits). With a Lego motor, change gears or battery power and measure the spin speed each time.

ENGINEERING BUILD

Make a digital flipbook

The digital version is the **main measuring tool** for the experiment, because a computer can show each drawing for an exact amount of time. The zoetrope's speed wobbles a little.

Step 1: Photograph the drawings

- Tape the phone to a stack of books, pointing straight down, so every photo is taken from the same spot.
- Tape a paper "corner guide" on the table so each drawing sits in exactly the same place.
- Use the same lighting for every photo, with no shadows from your hand.
- Name the photos in order: `frame01, frame02, … frame12`.

Step 2: Play them back at an exact FPS (pick one tool)

Stop Motion Studio

A free app for phone or tablet. Import the photos, then open the movie settings and set the frame rate. Export one video per speed, for example `car_4fps.mp4`.

Scratch (coding!)

Upload each drawing as a *costume* on one sprite. Then build:

```
when green flag clicked
forever
  next costume
  wait (1 / FPS) seconds
```

Make FPS a variable with a slider.

FPS tester (online guide)

The web version of this guide has a tester. Upload the photos, click a speed, or use **Mystery mode**, which hides the speed from volunteers.

darshangr.github.io/AI-understanding/science-fair/frame-rate-project.html

Step 3: Choose your test speeds

Most screens refresh **60 times per second**. Frame rates that divide evenly into 60 give each drawing the same amount of screen time. Those are the fairest speeds to test:

FPS	2	4	6	10	12	15	20	30
Each drawing is shown for	0.50 s	0.25 s	0.17 s	0.10 s	0.083 s	0.067 s	0.05 s	0.033 s
Screen refreshes per drawing	30	15	10	6	5	4	3	2

8 and 24 FPS don't divide evenly into 60, so some drawings stay on screen a little longer than others. That causes a tiny stutter, called *judder*. You can still test them. Movies at 24 FPS have the same problem on TVs, and the fix is called "3:2 pulldown." That makes a great fun fact for the board.

MAKE IT A FAIR TEST: DON'T GO SLOW-TO-FAST EVERY TIME

If speeds always go from slow to fast, volunteers can guess the pattern and say "smooth" too early. Use one of these:

- **Mystery order:** shuffle the speeds for each volunteer. You can write each speed on a card and draw cards from a hat.
- **Up and down:** go slow to fast once, then fast to slow once, and average the two thresholds.

THE ART PART

Draw the car chase

The movie is a **12-frame loop**: after frame 12 it goes straight back to frame 1 and keeps repeating. A loop is perfect for both the zoetrope and the flipbook.

Story idea: "The Getaway"

A red race car speeds along a road, with a police car close behind. The road lines and a tree slide backward to show speed. The wheels spin, and the race car bounces a little over bumps.

Frame	Race car	Police car	Background
1	Middle of the frame, wheels with spoke pointing up	Just entering at the left	Road dashes at start position, tree at the right edge
2–3	Same spot, spokes turn ⅛ of a turn each frame	Creeps forward a little	Dashes and tree slide left a little each frame
4–6	Hits a bump: body rises in 4, at the top in 5, lands in 6	Lights flash: on in odd frames, off in even frames	Keep sliding left
7–9	Pulls slightly ahead	Falls back a little	Tree reaches the left edge
10–12	Back to the middle, ready to loop to frame 1	Back to the frame 1 spot	Dashes back to the start, ready to loop

ANIMATOR TRICKS

- **Trace** the last drawing on a window or bright screen, then change just a little.
- **Small steps**: move things only about ¼ car-length per frame. Big jumps look choppy even at high FPS. That's an extension experiment on page 11!
- Draw the road line in exactly the same place on every frame so it doesn't wobble.
- Use **thick, dark outlines**. They show up best through zoetrope slits.

CHECK YOUR LOOP

Stack the 12 drawings and flip through them with your thumb like a flipbook. If something jumps between frame 12 and frame 1, redraw frame 12 so it leads smoothly back into frame 1.

Zoetrope picture strip template (12 boxes; cut out, tape the two halves end to end, and fit inside the drum)



The dashed line is the road, so keep it in the same place in every box. For an 8-inch drum, print this page at 100% and each box will be about 2 inches wide, which fits well.

RUN IT

Experiment procedure

Set up (same for every volunteer)

1. Use the same room, with the lights at the same brightness each time.
2. Seat the volunteer about **2 feet** from the screen. Put a tape mark on the floor for the chair.
3. If they wear glasses for distance or reading screens, have them wear them, and write it on the data sheet.
4. Prepare a **shuffled list of speeds** for this volunteer: 2, 4, 6, 8, 10, 12, 15, 20, 24, 30 FPS.

The script (read the same words every time)

> "I'm going to show you a short cartoon at different speeds. Each one plays for 5 seconds. After each one, tell me: did it look like **separate pictures** or like a **movie**? Then give it a smoothness score from 1 to 5 using this card. There are no right or wrong answers."

For each speed

1. Play the animation for **5 seconds**. Use a timer.
2. Ask: *"Pictures or movie?"* Write **P** or **M**.
3. Ask: *"Smoothness score, 1 to 5?"* Write the number.
4. Give them a 3-second break by looking away, then do the next speed.

After all speeds

1. Find this volunteer's **movie threshold**: the lowest FPS where they said **M** and gave a score of 4 or 5.
2. Optional zoetrope round: let them watch the zoetrope at 2 or 3 turntable speeds and score it the same way. Does the zoetrope look smoother than the screen at the same FPS? (The dark gap from each slit can help!)
3. Thank them!

Write down for each volunteer

- A code like V1, V2 or V3 instead of their real name
- Age group: under 10, 10–17, 18–49, 50+
- Glasses or contacts: yes or no
- Plays video games often: yes or no (fun extra!)

GOOD-SCIENTIST HABITS

- Don't hint, smile or say "good" when they answer.
- Don't let volunteers watch each other's turns.
- Test yourself last, because you know the animation too well. Mark your result as "the scientist."
- Write down anything surprising in a notes column.

COLLECT DATA

Data sheet

In each box, write **P** or **M** plus a score from 1 to 5 (example: M4). Print more copies as needed.

Vol.	Age group	Glasses?	2	4	6	8	10	12	15	20	24	30	Movie threshold (FPS)
V1													
V2													
V3													
V4													
V5													
V6													
V7													
V8													
V9													
V10													
V11													
V12													
V13													
V14													
V15													
Average score													
# who said "M"													

Column headings 2–30 are the frame rates in FPS.

Zoetrope round (optional)

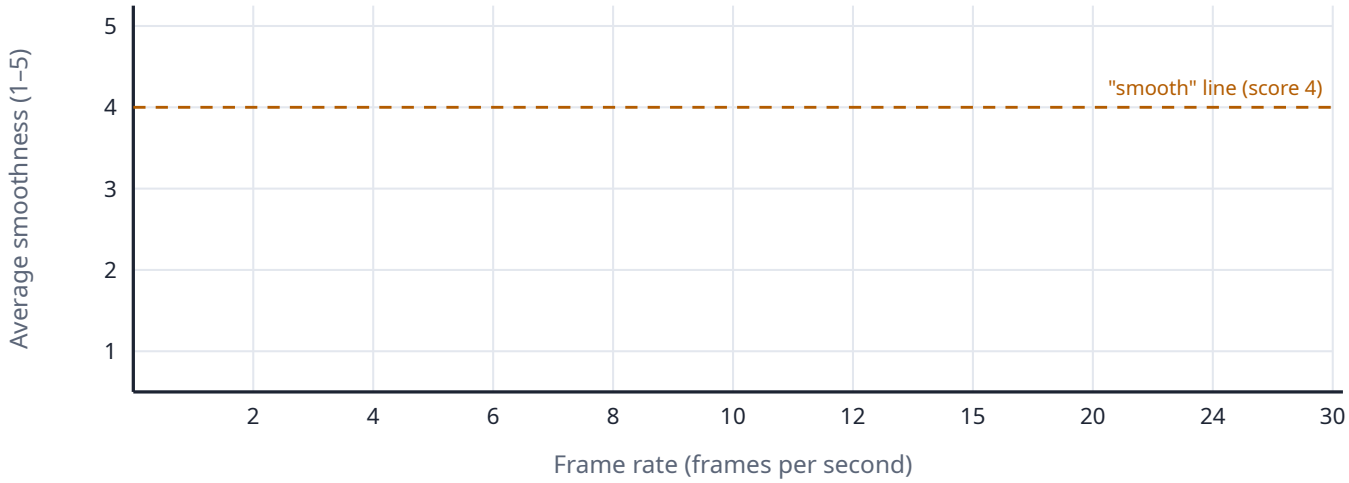
Spinner setting	Spins in 10 s (avg of 3)	Spins per second	Drawings	FPS	Average score

Notes and surprises

MAKE SENSE OF IT

Graphs and analysis

Graph 1: Average smoothness score at each frame rate (line graph)



Plot the "Average score" row from the data sheet, then join the dots. **Where does the line cross the dashed "smooth" line?** That's the group's movie threshold.

Graph 2: How many people had each movie threshold (bar graph)

Along the bottom, list each FPS. Draw a bar showing how many volunteers had that FPS as their threshold. Is there one tall bar (people agree), or are the bars spread out (people's eyes and brains differ)?

Do the math

Average (mean) threshold = add up all the thresholds ÷ the number of volunteers

= _____ ÷ _____ = _____ FPS

Range = highest threshold − lowest threshold

= _____ − _____ = _____ FPS

Questions to think about

- Was there a speed where almost *everyone* switched from "pictures" to "movie"?
- Did scores keep going up after 24 FPS, or did they level off? (A curve that flattens out is called a **plateau**.)
- Did kids and adults differ? People with and without glasses? Gamers and non-gamers?
- Did the zoetrope look smoother or choppier than the screen at about the same FPS? Why might that be?

WRAP IT UP

Conclusion and extensions

Conclusion (fill in the blanks)

I wanted to find out the lowest frame rate where drawings look like a movie. I tested _____ volunteers at _____ different speeds. The average movie threshold was _____ FPS, and the range was _____ to _____ FPS. My hypothesis was _____ (supported / not supported) because _____. This happens because the brain's motion detectors fill in the movement between frames (the phi phenomenon), and persistence of vision hides the gaps between them. One thing that surprised me was _____. If I did this again, I would _____.

Sources of error to mention

- Volunteers might guess what you want to hear, which is why you shuffle the speeds and use a script.
- The zoetrope's speed changes slightly as it spins.
- Screens can't show some frame rates perfectly evenly (judder at 8 and 24 FPS).
- A small number of volunteers: more people give more trustworthy averages.

Extension experiments (pick one if there's time)

Big steps vs. small steps

Make a second animation where the car moves **twice as far** each frame. Does it need a *higher* FPS to look smooth? (Prediction: yes. Smoothness depends on both FPS *and* how far things jump.)

Center vs. side vision

Have volunteers look at a dot to the *side* of the screen. The edges of the retina have more rods, which are very sensitive to flicker and motion. Does the animation look choppier out of the corner of the eye?

Bright vs. dim

Repeat the test with the screen brightness turned way down. Eyes respond more slowly in dim light. Does the threshold change?

Flicker fusion (micro:bit or Arduino)

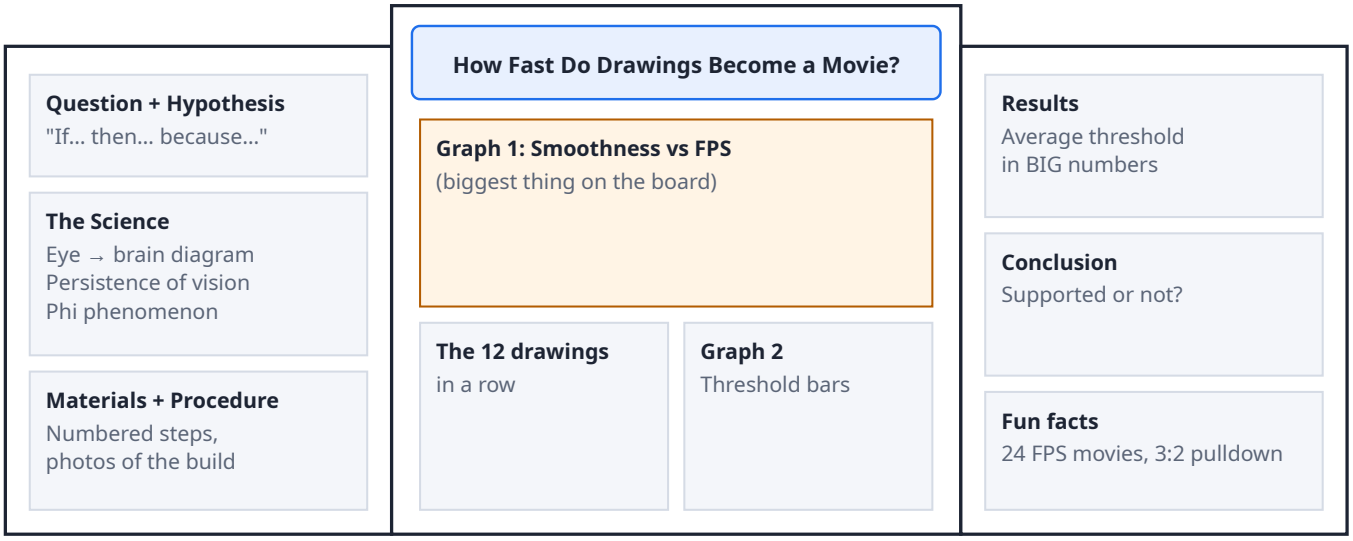
Blink an LED faster and faster. At what speed does it stop flickering and look steady? For most people it's around 50–60 flashes per second, which is much faster than the movie threshold. Why are they different?

EYE-DOCTOR CONNECTION

Eye doctors and vision scientists use flicker and motion tests to check how well the retina, optic nerve and brain are working together. Some eye conditions make it harder to see flicker or motion. That makes this project a real (simple) version of a vision test.

SHOW IT OFF

Display board plan



On the table in front: the spinning zoetrope + a tablet playing the flipbook (visitors can try it!)

Be ready to answer judges

- Why did you shuffle the order of speeds?
- What's the difference between persistence of vision and the phi phenomenon?
- Why do zoetropes need slits?
- Why are movies 24 FPS if your threshold was lower (or higher)?
- What would you test next?

Vocabulary

Frame
One still picture in an animation.

Frame rate (FPS)
How many frames are shown each second.

Persistence of vision
A fading trace of an image that lasts a split second after it's gone.

Phi phenomenon
The brain seeing motion between quickly flashed still pictures.

Threshold
The point where something changes, here from "pictures" to "movie."

Zoetrope
A spinning drum with slits that shows a looping animation.

Retina
The light-sensing layer at the back of the eye.

FOR GROWN-UPS

Notes for parents and helpers

Suggested schedule

When	Tasks
Week 1	Read the background page together and try the afterimage trick. Write the hypothesis. Plan the 12-frame story and sketch rough drawings.
Week 2	Finish the 12 final drawings. Photograph them and set up the digital flipbook. Build the zoetrope and measure its spin speeds.
Week 3	Test 10–20 volunteers (about 5 minutes each). Fill in the data sheet.
Week 4	Work out averages, draw graphs, write the conclusion and build the display board. Practice explaining the project out loud.

What results to expect (keep this to yourself until after testing, so it doesn't bias the hypothesis)

- Most people start to see *some* motion around **6–10 FPS**, and rate hand-drawn animation as smooth somewhere around **12–20 FPS**.
- Scores usually **plateau** above about 20–24 FPS. Hand-drawn animation already looks "finished" there, which is part of why 24 FPS became the movie standard.
- The threshold shifts a lot with **how far objects move per frame**. Small steps look smooth at lower FPS. If results are surprisingly high, the drawings may have big jumps, and that is itself a great finding to discuss.
- Differences between kids and adults, or gamers and non-gamers, are usually small with only 10–20 people. "No clear difference" is a perfectly valid result.

Where you can help

- All craft-knife cutting and any work near a spinning wheel or motor.
- Setting up the app or Scratch project the first time, then let him run it.
- Recruiting volunteers (neighbors, family, classmates at pickup).
- Checking the math on averages, but let him do it first.

> **CHECK THE FAIR'S RULES**
> Many school and regional science fairs treat any project with **human volunteers** as a "human subjects" project. That can require a simple consent form or teacher approval *before* testing starts. Ask the teacher early. Use codes (V1, V2...) instead of names on the board, and skip anyone who is sensitive to flashing images.

Helpful search terms

"how to make a zoetrope" · "phi phenomenon demonstration" · "beta movement animation" · "critical flicker fusion" · "Eadweard Muybridge horse in motion" (the famous 1878 photos that helped invent movies; a great addition to the board) · "Scratch animation costumes tutorial"